

Analysing E-Learning and Purchases using MySQL

Objectives:

To design and manage a relational **MySQL database** for an online learning platform and analyze learner purchase data to generate meaningful business insights. The project aims to understand **sales trends, learner purchasing behavior, course performance, and popular categories** using SQL queries, joins, aggregate functions, and data analysis techniques.

Dataset Description:

You will work with the following three tables:

Table: learners

Column	Description
learner_id	Primary Key
full_name	Learner name
country	Country of residence

Table: courses

Column	Description
course_id	Primary Key
course_name	Course title
category	Course category
unit_price	Course category

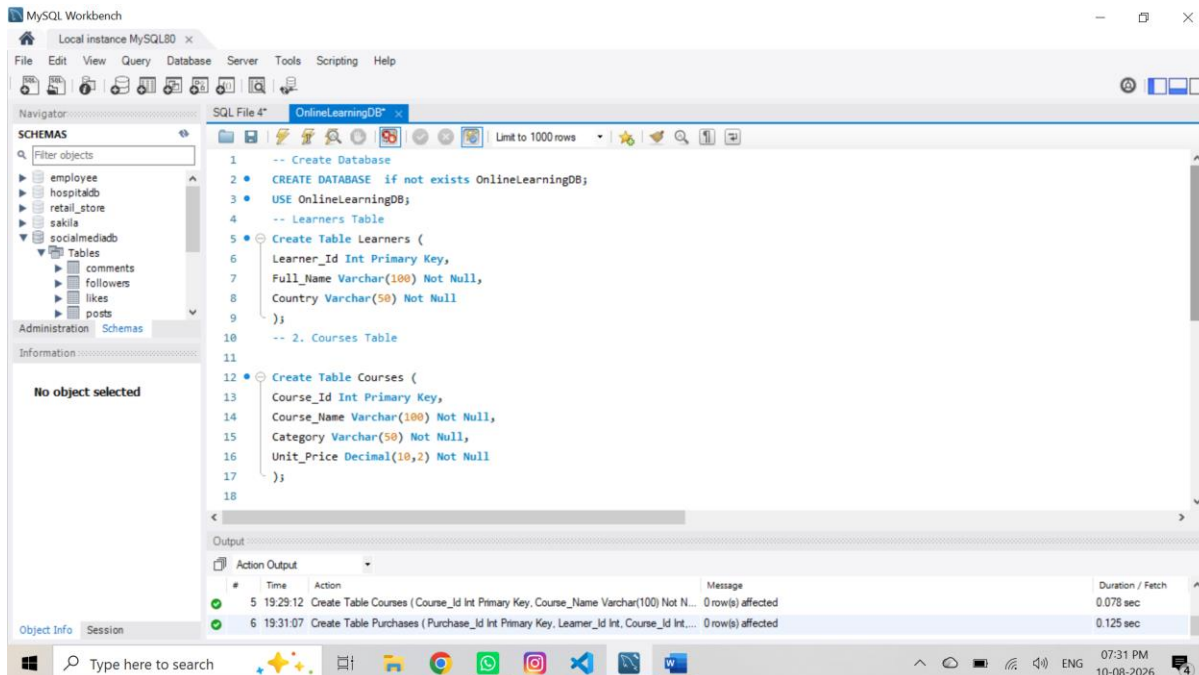
Table: purchases

Columns	Description
purchase_id	Primary Key
learner_id	Foreign Key → learners
course_id	Foreign Key → learners
quantity	Number of courses purchased
purchase_date	Date of purchase

1.Database Setup & Data Entry

Create a new database.

- Create the three tables with proper:
 - Primary keys, Foreign keys, Appropriate data types

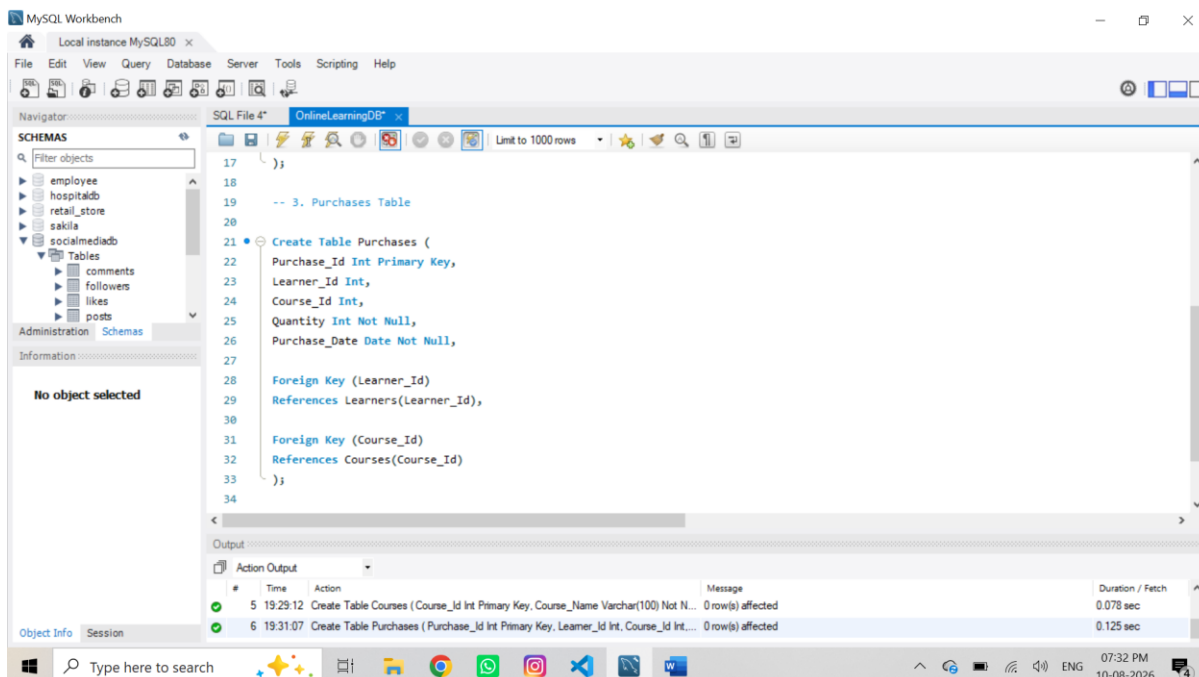


The screenshot shows the MySQL Workbench interface with the 'OnlineLearningDB' database selected. The SQL editor contains the following code:

```
1  -- Create Database
2  CREATE DATABASE IF NOT EXISTS OnlineLearningDB;
3  USE OnlineLearningDB;
4  -- Learners Table
5  CREATE TABLE Learners (
6  Learner_Id INT PRIMARY KEY,
7  Full_Name VARCHAR(100) NOT NULL,
8  Country VARCHAR(50) NOT NULL
9  );
10 -- 2. Courses Table
11
12 CREATE TABLE Courses (
13 Course_Id INT PRIMARY KEY,
14 Course_Name VARCHAR(100) NOT NULL,
15 Category VARCHAR(50) NOT NULL,
16 Unit_Price DECIMAL(10,2) NOT NULL
17 );
18
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
5	19:29:12	Create Table Courses (Course_Id Int Primary Key, Course_Name Varchar(100) Not N...	0 row(s) affected	0.078 sec
6	19:31:07	Create Table Purchases (Purchase_Id Int Primary Key, Learner_Id Int, Course_Id Int,...	0 row(s) affected	0.125 sec



The screenshot shows the MySQL Workbench interface with the 'OnlineLearningDB' database selected. The SQL editor contains the following code:

```
17  );
18
19  -- 3. Purchases Table
20
21  CREATE TABLE Purchases (
22  Purchase_Id INT PRIMARY KEY,
23  Learner_Id INT,
24  Course_Id INT,
25  Quantity INT NOT NULL,
26  Purchase_Date DATE NOT NULL,
27
28  FOREIGN KEY (Learner_Id)
29  REFERENCES Learners(Learner_Id),
30
31  FOREIGN KEY (Course_Id)
32  REFERENCES Courses(Course_Id)
33  );
34
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
5	19:29:12	Create Table Courses (Course_Id Int Primary Key, Course_Name Varchar(100) Not N...	0 row(s) affected	0.078 sec
6	19:31:07	Create Table Purchases (Purchase_Id Int Primary Key, Learner_Id Int, Course_Id Int,...	0 row(s) affected	0.125 sec

● Insert sample data:

- 4–5 learners
- 4–5 courses (different categories)
- 6–8 purchase records

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- employee
- hospitaldb
- retail_store
- sakila
- socialmediadb
 - Tables
 - comments
 - followers
 - likes
 - posts

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB x

Limit to 1000 rows

```
35 INSERT INTO learners
36 (learner_id, full_name, country)
37 VALUES
38 (1, 'Arun Kumar', 'India'),
39 (2, 'Priya sathiya', 'India'),
40 (3, 'Kalai Maha', 'USA'),
41 (4, 'Kathivel', 'Spain'),
42 (5, 'Tittu', 'USA'),
43 (6, 'Anitha Raj', 'India'),
44 (7, 'Karthik Kumar', 'India'),
45 (8, 'Robert jo', 'USA'),
46 (9, 'Sofia Karthick', 'Spain'),
47 (10, 'Vijay Jones', 'UK'),
48 (11, 'Divya Priya', 'India'),
49 (12, 'Daniel Wilson', 'USA'),
50 (13, 'Emma vel', 'UK'),
51 (14, 'Mala', 'France'),
52 (15, 'Aisha ', 'India');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	19:53:04	Insert into Courses (Course_Id, Course_Name, Category, Unit_Price) Values (101, 'S...	15 row(s) affected Records: 15 Duplicates: 0 Warnings: 0	0.016 sec
8	19:53:04	insert into purchases (purchase_id, learner_id, course_id, quantity, purchase_date) v...	18 row(s) affected Records: 18 Duplicates: 0 Warnings: 0	0.015 sec

Object Info Session

Type here to search

07:54 PM 10-08-2026

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- employee
- hospitaldb
- retail_store
- sakila
- socialmediadb
 - Tables
 - comments
 - followers
 - likes
 - posts

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB x

Limit to 1000 rows

```
55 -- Insert Courses
56 Insert Into Courses
57 (Course_Id, Course_Name, Category, Unit_Price)
58 Values
59 (101, 'Sql Basics', 'Beginner', 3000.00),
60 (102, 'Excel Analytics', 'Beginner', 4000.00),
61 (103, 'Power BI', 'Intermediate', 6000.00),
62 (104, 'Python For Data Analysis', 'Advanced', 9000.00),
63 (105, 'Machine Learning', 'Advanced', 12000.00),
64 (106, 'Advanced Excel', 'Beginner', 4500.00),
65 (107, 'SQL Advanced', 'Intermediate', 5500.00),
66 (108, 'Power BI Advanced', 'Intermediate', 7500.00),
67 (109, 'Python Programming', 'Advanced', 8500.00),
68 (110, 'Data Science', 'Advanced', 10000.00),
69 (111, 'Statistics for Data Analysis', 'Beginner', 5000.00),
70 (112, 'Tableau Analytics', 'Intermediate', 6500.00),
71 (113, 'Deep Learning', 'Advanced', 15000.00),
72 (114, 'Data Visualization', 'Beginner', 3500.00),
```

Output

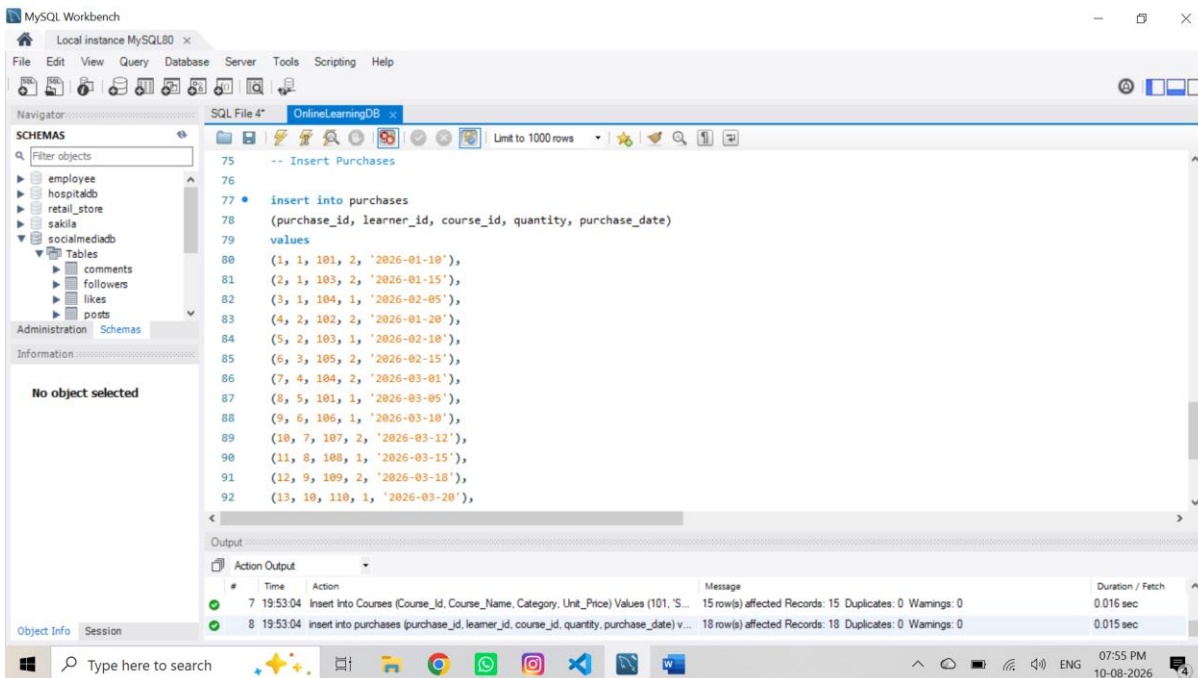
Action Output

#	Time	Action	Message	Duration / Fetch
7	19:53:04	Insert Into Courses (Course_Id, Course_Name, Category, Unit_Price) Values (101, 'S...	15 row(s) affected Records: 15 Duplicates: 0 Warnings: 0	0.016 sec
8	19:53:04	insert into purchases (purchase_id, learner_id, course_id, quantity, purchase_date) v...	18 row(s) affected Records: 18 Duplicates: 0 Warnings: 0	0.015 sec

Object Info Session

Type here to search

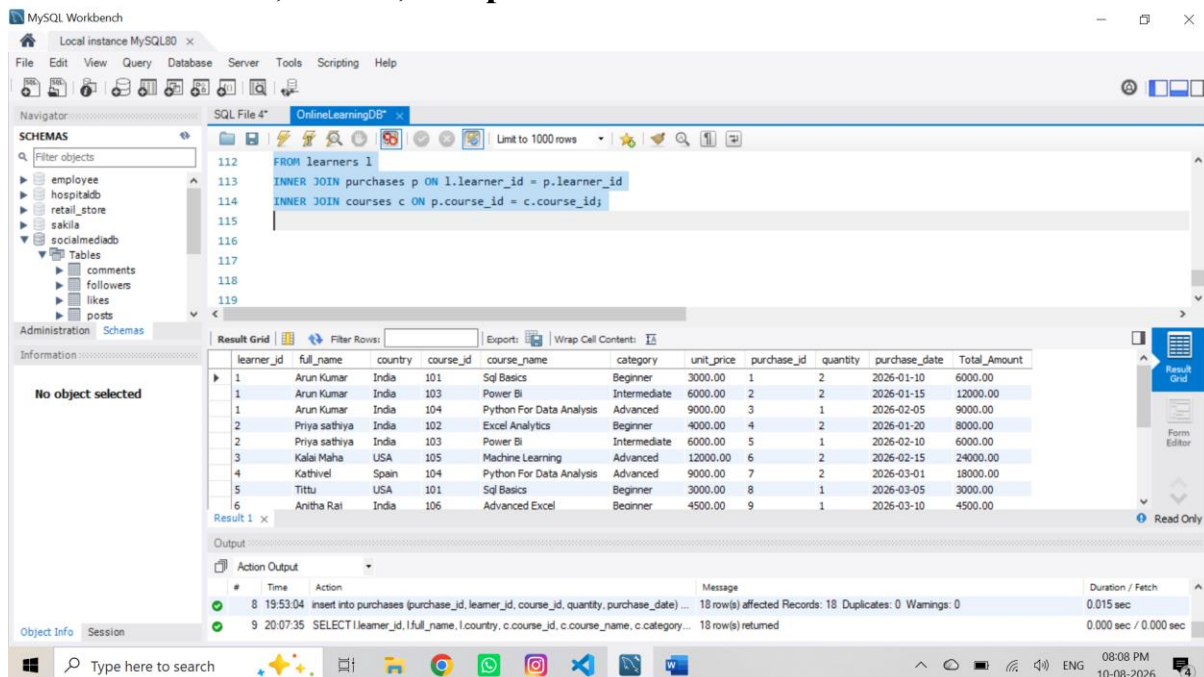
07:55 PM 10-08-2026



1. Data Exploration Using Joins

Use INNER, LEFT, and RIGHT JOIN to:

- Combine learner, course, and purchase data



Display: Learner name, Course name, Category, Quantity, Total amount, Purchase date

MySQL Workbench interface showing a SQL query and its result grid. The query is a JOIN between learners, purchases, and courses. The result grid shows columns: Learner_Name, Course_Name, Category, Quantity, Total_Amount, Purchase_Date. A Google Drive notification is visible on the right.

```

121 c.category AS Category,
122 p.quantity AS Quantity,
123 (p.quantity * c.unit_price) AS Total_Amount,
124 p.purchase_date AS Purchase_Date
125 FROM learners l
126 INNER JOIN purchases p ON l.learner_id = p.learner_id
127 INNER JOIN courses c ON p.course_id = c.course_id;
128

```

Learner_Name	Course_Name	Category	Quantity	Total_Amount	Purchase_Date
Arun Kumar	Sql Basics	Beginner	2	6000.00	2026-01-10
Arun Kumar	Power BI	Intermediate	2	12000.00	2026-01-15
Arun Kumar	Python For Data Analysis	Advanced	1	9000.00	2026-02-05
Priya sathya	Excel Analytics	Beginner	2	8000.00	2026-01-20
Priya sathya	Power BI	Intermediate	1	6000.00	2026-02-10
Kalai Maha	Machine Learning	Advanced	2	24000.00	2026-02-15
Kathirvel	Python For Data Analysis	Advanced	2	18000.00	2026-03-01
Tittu	Sql Basics	Beginner	1	3000.00	2026-03-05
Anitha Raj	Advanced Excel	Beginner	1	4500.00	2026-03-10

Presentation Requirements:

- Format currency to 2 decimal places
- Use column aliases
- Sort by the highest total amount

MySQL Workbench interface showing a modified SQL query with formatting and sorting. The result grid is sorted by Total_Amount in descending order. The query uses FORMAT for currency and column aliases.

```

138 p.quantity AS Quantity,
139 FORMAT(p.quantity * c.unit_price, 2) AS Total_Amount,
140 p.purchase_date AS Purchase_Date
141 from learners l
142 Left Join purchases p ON l.learner_id = p.learner_id
143 Left Join courses c ON p.course_id = c.course_id
144 order by Total_Amount DESC;
145

```

Learner_Name	Course_Name	Category	Quantity	Total_Amount	Purchase_Date
Arun Kumar	Python For Data Analysis	Advanced	1	9,000.00	2026-02-05
Priya sathya	Excel Analytics	Beginner	2	8,000.00	2026-01-20
Robert jo	Power BI Advanced	Intermediate	1	7,500.00	2026-03-15
Daniel Wilson	Tableau Analytics	Intermediate	1	6,500.00	2026-03-25
Arun Kumar	Sql Basics	Beginner	2	6,000.00	2026-01-10
Priya sathya	Power BI	Intermediate	1	6,000.00	2026-02-10
Anitha Raj	Advanced Excel	Beginner	1	4,500.00	2026-03-10
Tittu	Sql Basics	Beginner	1	3,000.00	2026-03-05
Kalai Maha	Machine Learning	Advanced	2	24,000.00	2026-02-15

Core Analytical Queries (Q1–Q5)

Q1. Display each learner's total spending with their country.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

hospitaldb

retail_store

sakila

socialmediadb

Tables

comments

followers

likes

posts

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB*

Limit to 1000 rows

```

153 join courses c ON p.course_id = c.course_id
154 group by
155     1.learner_id,
156     1.full_name,
157     1.country
158 order by Total_Spending DESC;
159
160

```

Result Grid

Filter Rows:

Exports

Wrap Cell Contents:

Learner_Name	Country	Total_Spending
Arun Kumar	India	27000.00
Kalai Maha	USA	24000.00
Kathirvel	Spain	18000.00
Sofia Karthick	Spain	17000.00
Emma vel	UK	15000.00
Priya sathya	India	14000.00
Aisha	India	14000.00
Karthik Kumar	India	11000.00
Mala	France	10500.00

Result 4 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	20:25:07	SELECT l.full_name AS Learner_Name, c.course_name AS Course_Name, c.categ...	18 row(s) returned	0.000 sec / 0.000 sec
12	20:31:37	select l.full_name AS Learner_Name, l.country AS Country, SUM(p.quantity * c.unit_...	15 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

08:31 PM 10-08-2026

Q2. Find the top 3 most purchased courses by quantity

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

hospitaldb

retail_store

sakila

socialmediadb

Tables

comments

followers

likes

posts

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB*

Limit to 1000 rows

```

163 c.course_name as Course_Name,
164 sum(p.quantity) as Total_Quantity
165 from Courses c
166 inner join purchases p on c.course_id=p.Course_id
167 group by c.course_id,c.course_name
168 order by Total_Quantity desc
169 limit 3;

```

Result Grid

Filter Rows:

Exports

Wrap Cell Contents:

Fetch rows:

Course_Name	Total_Quantity
Sql Basics	3
Power Bi	3
Python For Data Analysis	3

Result 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	20:40:53	select c.course_name as Course_Name, sum(p.quantity) as Total_Quantity from Co...	Error Code: 1146. Table 'onlinelearningdb.purchase' doesn't exist	
14	20:41:25	select c.course_name as Course_Name, sum(p.quantity) as Total_Quantity from Co...	3 row(s) returned	

Object Info Session

Type here to search

08:41 PM 10-08-2026

Q3. Show each category's:

- Total revenue
- Number of unique learners

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee
hospitaldb
retail_store
sakila
socialmediadb
sys
world

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB

Limit to 1000 rows

```

173 -- Number of unique learners
174 SELECT
175 c.category AS Category,
176 SUM(p.quantity * c.unit_price) AS Total_Revenue,
177 COUNT(DISTINCT p.learner_id) AS Unique_Learners
178 FROM courses c
179 INNER JOIN purchases p ON c.course_id = p.course_id
180 GROUP BY c.category

```

Result Grid

Category	Total_Revenue	Unique_Learners
Advanced	107000.00	7
Intermediate	43000.00	5
Beginner	42000.00	6

Result 1 Result 2 Result 3 Result 4 Result 5 Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	11:03:17	select c.course_name as Course_Name, sum(p.quantity) as Total_Quantity from Co...	3 row(s) returned	0.000 sec / 0.000 sec
14	11:03:17	SELECT c.category AS Category, SUM(p.quantity * c.unit_price) AS Total_Revenu...	3 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

11:03 AM 11-08-2026

Q4. List learners who purchased from more than one category.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee
hospitaldb
retail_store
sakila
socialmediadb
sys
world

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB

Limit to 1000 rows

```

186 l.learner_id,
187 l.full_name
188 from learners l
189 inner join purchases p on l.learner_id = p.learner_id
190 inner join courses c on p.course_id = c.course_id
191 group by l.learner_id, l.full_name
192 having count(distinct c.category) > 1
193

```

Result Grid

learner_id	full_name
1	Arun Kumar
2	Priya sathya

Result 7 x

Output

Action Output

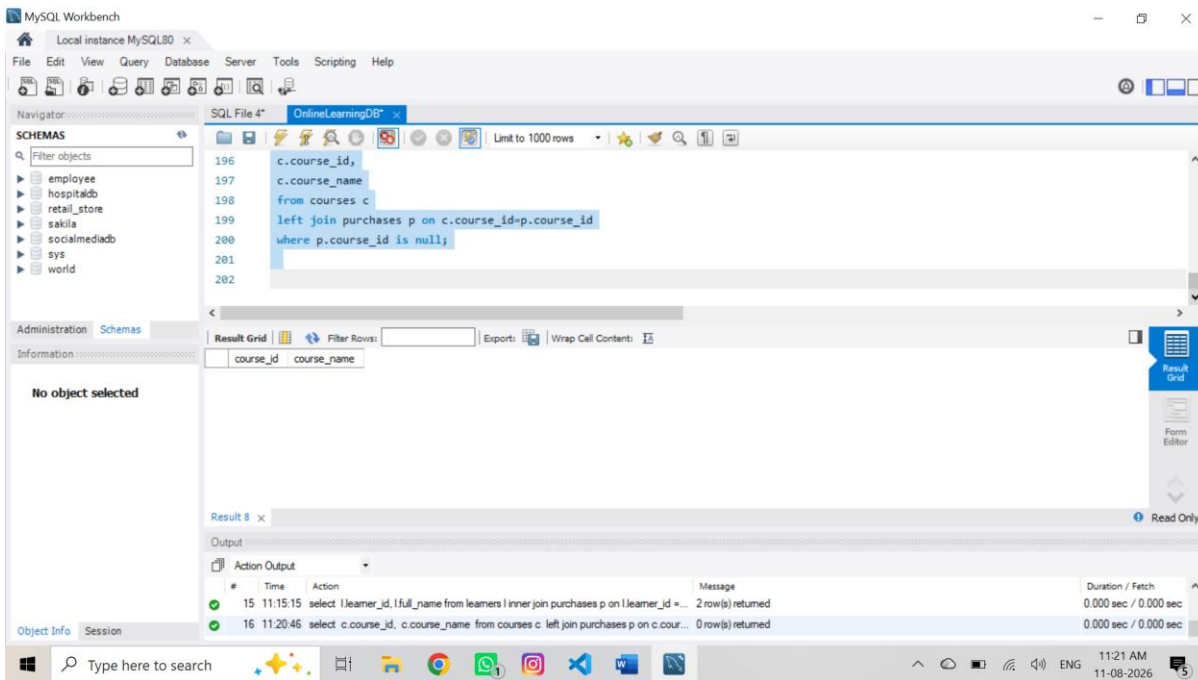
#	Time	Action	Message	Duration / Fetch
14	11:03:17	SELECT c.category AS Category, SUM(p.quantity * c.unit_price) AS Total_Revenu...	3 row(s) returned	0.000 sec / 0.000 sec
15	11:15:15	select l.learner_id, l.full_name from learners l inner join purchases p on l.learner_id = ...	2 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

11:15 AM 11-08-2026

Q5. Identify courses never purchased.



MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee
hospitaldb
retail_store
sakila
socialmediadb
sys
world

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB

Limit to 1000 rows

217
218 -- Q7. Display courses whose price is higher than any course in the 'Beginner' category
219
220 select
221 course_id,
222 unit_price
223 from courses
224 where unit_price > any(select unit_price

Result Grid

course_id	course_name	unit_price
102	Excel Analytics	4000.00
103	Power BI	6000.00
104	Python For Data Analysis	9000.00
105	Machine Learning	12000.00
106	Advanced Excel	4500.00
107	SQL Advanced	5500.00
108	Power BI Advanced	7500.00
109	Python Programming	8500.00
110	Data Science	10000.00

courses 10 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	11:26:53	Select L.Learner_Id, L.Full_Name, Sum(P.Quantity * C.Unit_Price) As Total_Spend...	7 row(s) returned	0.000 sec / 0.000 sec
18	11:31:58	select course_id, course_name, unit_price from courses where unit_price > any(sele...	14 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

11:34 AM 11-08-2026

Q8 . Find learners who spent more than the average spending in their country.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee
hospitaldb
retail_store
sakila
socialmediadb
sys
world

Administration Schemas

Information

No object selected

SQL File 4* OnlineLearningDB

Limit to 1000 rows

239
240 having sum(p.quantity * c.unit_price) > (select avg(country_spending)
241 from (select l2.country, sum(p2.quantity * c2.unit_price) as country_spending
242 from learners l2
243 inner join purchases p2 on l2.learner_id = p2.learner_id
244 inner join courses c2 on p2.course_id = c2.course_id
245 group by l2.country, l2.learner_id) as country_avg
246

Result Grid

learner_id	full_name	country	total_spending
1	Arun Kumar	India	27000.00
2	Priya saithya	India	14000.00
3	Kalek Maha	USA	24000.00
4	Kathivel	Spain	18000.00
9	Sofia Karthick	Spain	17000.00
13	Emma vel	UK	15000.00
15	Aisha	India	14000.00

Result 11 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
18	11:31:58	select course_id, course_name, unit_price from courses where unit_price > any(sele...	14 row(s) returned	
19	11:41:15	select l.learner_id, l.full_name, l.country, sum(p.quantity * c.unit_price) as total_spe...	7 row(s) returned	

Object Info Session

Type here to search

OneDrive

Screenshot saved
The screenshot was added to OneDrive.

11:41 AM 11-08-2026

5. CTE, CASE, View, and NULL Handling

**Q9. Use a CTE to calculate total spending per learner, then:
Display learners with spending above 10,000.**

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```

group by l.learner_id, l.full_name)
select
  learner_id,
  full_name,
  total_spending
from learner_spending
where total_spending > 10000;

```

The Results Grid displays the following data:

learner_id	full_name	total_spending
1	Arun Kumar	27000.00
2	Priya sathya	14000.00
3	Kalai Maha	24000.00
4	Kathivel	18000.00
7	Karthik Kumar	11000.00
9	Sofia Karthick	17000.00
13	Emma vel	15000.00
14	Mala	10500.00
15	Aisha	14000.00

The Output tab shows the execution message: "7 row(s) returned" and "0.000 sec / 0.000 sec".

Q10. CASE Expression Classify learners based on spending:

- Above 15,000 → “High Value”,
- 8,000–15,000 → “Medium Value”,
- Below 8,000 → “Low Value”.

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following query:

```

when sum(p.quantity * c.unit_price) >= 8000 then 'medium value'
else 'low value'
end as spending_category
from learners l
join purchases p on l.learner_id = p.learner_id
join courses c on p.course_id = c.course_id
group by l.learner_id, l.full_name;

```

The Results Grid displays the following data:

learner_id	full_name	total_spending	spending_category
1	Arun Kumar	27000.00	high value
2	Priya sathya	14000.00	medium value
3	Kalai Maha	24000.00	high value
4	Kathivel	18000.00	high value
5	Tittu	3000.00	low value
6	Anitha Raj	4500.00	low value
7	Karthik Kumar	11000.00	medium value
8	Robert jo	7500.00	low value
9	Sofia Karthick	17000.00	high value

The Output tab shows the execution message: "15 row(s) returned" and "0.000 sec / 0.000 sec".

Q11 . NULL Handling

- Display all courses and replace NULL purchase counts with 0 using: IFNULL() or COALESCE()

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

hospitaldb

retail_store

sakila

socialmediadb

sys

world

SQL File 4* OnlineLearningDB

Limit to 1000 rows

287 c.course_id,

288 c.course_name,

289 ifnull(count(p.purchase_id), 0) as purchase_count

290 from courses c

291 left join purchases p on c.course_id = p.purchase_id

292 group by c.course_id, c.course_name;

293

294

Administration Schemas

Information

No object selected

Result Grid

course_id	course_name	purchase_count
101	Sql Basics	2
102	Excel Analytics	1
103	Power BI	2
104	Python For Data Analysis	2
105	Machine Learning	1
106	Advanced Excel	1
107	SQL Advanced	1
108	Power BI Advanced	1
109	Python Programming	1

Result 14 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	11:59:47	select l.learner_id, l.full_name, sum(p.quantity * c.unit_price) as total_spending, cas...	15 row(s) returned	0.000 sec / 0.000 sec
22	12:04:57	select c.course_id, c.course_name, ifnull(count(p.purchase_id), 0) as purchase_co...	15 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Type here to search

12:05 PM 11-08-2026

Q12 . View

- Create a view:

category_performance_view

- Showing
- Category
- Total revenue
- Number of purchases
- Average revenue per purchase

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

employee

hospitaldb

retail_store

sakila

socialmediadb

sys

world

SQL File 4* OnlineLearningDB

Limit to 1000 rows

307 count(p.purchase_id) as number_of_purchases,

308 avg(p.quantity * c.unit_price) as average_revenue_per_purchase

309 from courses c

310 join purchases p on c.course_id = p.purchase_id

311 group by c.category;

312

313 select * from category_performance_view;

314

Administration Schemas

Information

No object selected

Result Grid

category	total_revenue	number_of_purchases	average_revenue_per_purchase
Beginner	42000.00	6	7000.000000
Intermediate	43000.00	5	8600.000000
Advanced	107000.00	7	15285.714286

category_performance_view 17 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
30	12:18:43	CREATE VIEW category_performance_view AS select c.category as category, su...	0 row(s) affected	0.062 sec
31	12:18:51	select from category_performance_view LIMIT 0, 1000	3 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Type here to search

12:18 PM 11-08-2026

Conclusion :

The online learning platform database was successfully designed and managed using MySQL. SQL queries, joins, aggregate functions, CASE expressions, CTEs, NULL handling, and views were used to analyze learner purchases. The analysis provided useful insights into sales trends, learner spending behavior, course performance, and popular categories, helping the platform make better data-driven business decisions.